## String Palindrome

```
String str = "madam";
boolean isPalindrome = IntStream.range(0, str.length() / 2)
                    .allMatch(i -> str.charAt(i) == str.charAt(str.length() - 1 - i));
System.out.println(isPalindrome ? "Palindrome" : "Not");
```

## Number palindrome

```
int num = 1271;
String str = String.valueOf(num);
boolean isPalindrome = IntStream.range(0, str.length() / 2)
                    .allMatch(i -> str.charAt(i) == str.charAt(str.length() - 1 - i));
System.out.println(isPalindrome ? "Palindrome" : "Not");
```

```
          int num = 1271;
          String str = String.valueOf(num);
          boolean isPalindrome = IntStream.range(0, str.length() / 2)
                    .allMatch(i -> str.charAt(i) == str.charAt(str.length() - 1 - i));
          System.out.println(isPalindrome ? "Palindrome" : "Not");
```

## String reverse

```
String name = "ankur";
          String reversedString = IntStream.range(0, name.length()).mapToObj(i -> name.charAt(str.length() - i))
                    .map(String::valueOf).collect(Collectors.joining());
```

```
System.out.println(reversedString);
```

## String anagram

```java
String str1 = "listen", str2 = "silent";

boolean isAnagram = str1
                        .chars()
                        .mapToObj(item -> String.valueOf((char) item))
                        .sorted()
                        .collect(Collectors.joining())
                        .equals(str2
                        .chars()
                        .mapToObj(item1 -> String.valueOf((char) item1))
                        .sorted()
                        .collect(Collectors.joining()));
System.out.println(isAnagram ? "Anagram" : "Not");
```

## Number Anagram

```
public class NumberAnagram {
public static boolean isAnagram(int num1, int num2) {
// Convert numbers to char arrays
char[] arr1 = String.valueOf(num1).toCharArray();
char[] arr2 = String.valueOf(num2).toCharArray();

// Sort using streams
//Option 1
String sorted1 = Arrays.stream(String.valueOf(num1).split(""))
.sorted()
.collect(StringBuilder::new, StringBuilder::append, StringBuilder::append).toString();

String sorted2 = Arrays.stream(String.valueOf(num2).split("")).sorted()
.collect(StringBuilder::new, StringBuilder::append, StringBuilder::append).toString();

//Option 2
String sorted1 = Arrays.stream(String.valueOf(num1).split("")).sorted().collect(Collectors.joining());
String sorted2 = Arrays.stream(String.valueOf(num2).split("")).sorted().collect(Collectors.joining());
return sorted1.equals(sorted2);
}

public static void main(String[] args) {
int num1 = 12345;
int num2 = 54321;
int num3 = 12435;
int num4 = 12344;
System.out.println(num1 + " & " + num2 + " are anagrams? " + isAnagram(num1, num2));
System.out.println(num1 + " & " + num3 + " are anagrams? " + isAnagram(num1, num3));
System.out.println(num1 + " & " + num4 + " are anagrams? " + isAnagram(num1, num4));
}
}
```

**IntelliJ Coding**

```java
public class NumberPalindrom {

    public static void main(String[] args) {
       int num = 1271;
       String str = String.valueOf(num);
       boolean isPalindrome = IntStream.range(0, str.length() / 2)
             .allMatch(i -> str.charAt(i) == str.charAt(str.length() - 1 - i));
       System.out.println(isPalindrome ? "Palindrome" : "Not");

       long count1 = IntStream.range(0, str.split("").length).count();
       System.out.println(count1);

       long count2 = Arrays.stream(str.split("")).count();
       System.out.println(count2);

       // reverse a String using streams
       String name = "ankur";
       String reversedString = IntStream.range(0, name.length()).mapToObj(i -> name.charAt(str.length() - i))
             .map(String::valueOf).collect(Collectors.joining());
       System.out.println(reversedString);

       // String anagram
       String str1 = "listen", str2 = "silent";

       boolean isAnagram = str1.chars().mapToObj(item -> String.valueOf((char) item)).sorted()
             .collect(Collectors.joining()).equals(str2.chars().mapToObj(item1 -> String.valueOf((char) item1))
                   .sorted().collect(Collectors.joining()));
       System.out.println(isAnagram ? "Anagram" : "Not");

       // Number anagram
       int num1 = 123, num2 = 321;
       char[] arr1 = String.valueOf(num1).toCharArray();
       String arr1Str=Arrays.stream(arr1)
```

```
                .sorted()
                .mapToObj(item->String.valueOf((char)item))
                .collect(Collectors.joining());
//      System.out.println(arr1Str);)

    }
}
```